Unit 1 Classwork — Real Numbers

Converting · Classification · Squares · Estimating Roots · Ordering

ANSWER KEY

A. Converting Fractions ↔ Decimals

Problem 1

Convert $\frac{3}{8}$ to a decimal.

0.375

Problem 2

Convert $\frac{5}{6}$ to a decimal (round to the thousandths, use a repeating bar notation if needed).

$0.8\overline{3}$

Problem 3

Convert 0.45 to a fraction in simplest form.

$\frac{9}{20}$

Problem 4

Convert $\frac{7}{20}$ to a decimal.

0.35

Problem 5

Convert $0.\overline{6}$ (repeating) to a fraction.

$\frac{2}{3}$

Problem 6

Convert $\frac{1}{3}$ to a decimal.

$0.\overline{3}$

B. Classifying Numbers

Problem 1

Is √16 rational or irrational?

rational

Problem 2
Is √7 rational or irrational?

irrational

Problem 3
Is 0.454545... (repeating) rational or irrational?

rational

Problem 4
Is π rational or irrational?

irrational

Problem 5
Is −5 a whole number, an integer, or both?

integer

Problem 6
Is √36 rational or irrational?

rational

C. Perfect Squares

Problem 1
What is 9^2?

81

Problem 2
What is 12^2?

144

Problem 3
What is √121?

11

Problem 4
What is √225?

15

D. Estimating Square Roots

Problem 1

Between which two consecutive integers does $\sqrt{50}$ fall?

7 and 8

Problem 2

Estimate $\sqrt{50}$ to the nearest tenth.

7.1

Problem 3

Between which two consecutive integers does $\sqrt{90}$ fall?

9 and 10

E. Ordering Rational & Irrational Numbers

Problem 1

Order from least to greatest: $\sqrt{10}$, 3, $\frac{10}{3}$

3, $\sqrt{10}$, $\frac{10}{3}$

Problem 2

Which is greater: $\sqrt{2}$ or $\frac{3}{2}$?

$\frac{3}{2}$

Problem 3

Which is greater: π or $\sqrt{10}$?

$\sqrt{10}$